Flood susceptibility and hazard zonation · Khagaria, North Bihar · five models, Max-CSI thresholds

Multi-mission SAR inventory
2008 · 2020 · 2021 (824 km² flooded)

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14 CgFs at 30 m · district mask
real slope from DEM · max VIF 1.8

↓

RF · XGBoost · CatBoost · ANN · DNN
70 / 10 / 20 split

↓

Max-CSI threshold on validation
metrics on independent test set

↓

Susceptibility + hazard maps
SHAP for all five models



